

An Aligned-Center Recursion for Products of q -Integers

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Abstract

Connelly, Ito, Martinez, Shevchenko, and Yang conjectured a sufficient condition for the unimodality of $\prod_{i=1}^k [a_i]_q [b]_{q^r}$, and separately conjectured that the condition is necessary when $k \leq 3$ or $r \leq 3$. We prove the sufficient condition for every $k \geq 1$ and $r \geq 2$. The source's example at $k = r = 4$ shows that necessity fails in general, so no necessity claim is made here. The proof is an induction based on the identity

$$[a + r]_q [b + 1]_{q^r} = q^r [a]_q [b]_{q^r} + [a + r(b + 1)]_q.$$

After multiplication by the remaining factors, the two terms on the right are symmetric unimodal about exactly the same center. Allocating the available increments among the parameters completes the induction.

1 Introduction

For a positive integer n , write

$$[n]_q = 1 + q + \cdots + q^{n-1}.$$

Products of such polynomials arise naturally in questions about q -Fibonomial coefficients. Connelly, Ito, Martinez, Shevchenko, and Yang formulated the following sufficient condition [1, Conjecture 5.4]: if $r \geq 2$, $k \geq 1$, and $a_1, \dots, a_k, b$ are positive integers satisfying

$$r \mid a_i \quad \text{for some } i, \quad \text{or} \quad b \leq 1 + \sum_{i=1}^k \left\lfloor \frac{a_i}{r} \right\rfloor, \quad (1)$$

then

$$P(q) = \prod_{i=1}^k [a_i]_q [b]_{q^r} \quad (2)$$

is unimodal. They also conjecture necessity when $k \leq 3$ or $r \leq 3$, prove several precursor cases, and verify bounded instances.

The sufficient direction of Conjecture 5.4 was open in general. We prove it for all $k \geq 1$ and $r \geq 2$. Necessity is a separate question: the source itself observes that $([3]_q)^4 [2]_{q^4}$ is unimodal although it violates (1) [1, after Conjecture 5.4], so no necessity claim is made here.

Theorem 1. *Let $r \geq 2$, $k \geq 1$, and let $a_1, \dots, a_k, b$ be positive integers. If either condition in (1) holds, then the polynomial (2) is symmetric and unimodal.*

The hypothesis $r \geq 2$ is inherited from the source's statement rather than used in the proof. Identity (3) and the argument remain valid for $r = 1$, where the divisibility condition is automatic.

2 Two elementary lemmas

A finite nonnegative coefficient sequence is *symmetric unimodal about C* if coefficients at exponents equidistant from C agree and the coefficients weakly increase up to C . We allow initial and terminal zeros; this convention makes translates convenient.

Lemma 2. *If F and G are nonnegative symmetric unimodal polynomials about centers C_1 and C_2 , respectively, then FG is symmetric unimodal about $C_1 + C_2$. A sum of nonnegative symmetric unimodal polynomials having a common center C is symmetric unimodal about C .*

Proof. The assertion about sums follows coefficientwise. For products, decompose each symmetric unimodal coefficient sequence into a nonnegative linear combination of indicator sequences of nested intervals having the same center. Unimodality permits zeros only before and after the nonzero support, not internally, which licenses this nested-interval decomposition. The convolution of two interval indicators is a symmetric trapezoidal sequence, hence unimodal, and all resulting convolutions have the sum of the original centers. Summing them proves the claim. $\square$

The next identity supplies the induction.

Lemma 3 (aligned-center recursion). *For positive integers a, b, r ,*

$$[a + r]_q [b + 1]_{q^r} = q^r [a]_q [b]_{q^r} + [a + r(b + 1)]_q. \quad (3)$$

Proof. Use

$$[a + r]_q = [r]_q + q^r [a]_q, \quad [b + 1]_{q^r} = 1 + q^r [b]_{q^r}.$$

After subtracting the first term on the right side of (3), the remaining terms are

$$\begin{aligned} [r]_q [b + 1]_{q^r} + q^r [a]_q (1 + (q^r - 1)[b]_{q^r}) &= [r(b + 1)]_q + q^{r(b+1)} [a]_q \\ &= [a + r(b + 1)]_q. \end{aligned}$$

For a coefficient-level check independent of this manipulation, the left side counts pairs (x, j) with $0 \leq x < a + r$ and $0 \leq j \leq b$, weighted by $x + rj$. The pairs with $x < a$ and $j \geq 1$ give $q^r [a]_q [b]_{q^r}$. The complementary pairs have either $j = 0$, or $x \geq a$ and $j \geq 1$; their weights occur once each and fill the consecutive interval from 0 to $a + r(b + 1) - 1$. They give the other term in (3). $\square$

3 Proof of the theorem

Suppose first that $r \mid a_i$ for some i . Writing $a_i = rs$, we have

$$[a_i]_q [b]_{q^r} = [r]_q [s]_{q^r} [b]_{q^r}.$$

The product $[s]_z [b]_z$ is symmetric unimodal in z . Substitution $z = q^r$, followed by multiplication by $[r]_q$, repeats each of its coefficients in a block of length r ; the resulting sequence remains symmetric unimodal. Lemma 2 then permits multiplication by the remaining factors $[a_j]_q$. This proves the divisibility branch.

Now assume no a_i is divisible by r , and that the inequality in (1) holds. Put $c_i = \lfloor a_i/r \rfloor$. Choose integers d_i such that

$$0 \leq d_i \leq c_i, \quad \sum_{i=1}^k d_i = b - 1. \quad (4)$$

Such a choice exists by (1). Define $a_i^{(0)} = a_i - rd_i$. Every $a_i^{(0)}$ is positive because $r \nmid a_i$. At spacer length one, the base polynomial is

$$\prod_{i=1}^k [a_i^{(0)}]_q,$$

which is symmetric unimodal by Lemma 2.

Starting from this base, perform the $b - 1$ increments allocated in (4). At one step, let a be the current value of the selected length, let c be the current spacer length, and let $B(q)$ be the product of all other current ordinary q -integers. Lemma 3 gives

$$[a + r]_q B(q) [c + 1]_{q^r} = q^r [a]_q B(q) [c]_{q^r} + [a + r(c + 1)]_q B(q). \quad (5)$$

By induction, the first term on the right is a translate of a symmetric unimodal polynomial. The second term is a product of ordinary q -integers, hence is symmetric unimodal.

It remains only to check their centers. If the old degree is

$$E = (a - 1) + \deg B + r(c - 1),$$

then the new degree is $E + 2r$. The support endpoints of the translated first term sum to $E + 2r$, while the degree of the second term is

$$a + r(c + 1) - 1 + \deg B = E + 2r.$$

Thus both summands in (5) have center $(E + 2r)/2$. Their sum is symmetric unimodal by Lemma 2. Iterating (5) reconstructs all final lengths a_i and the final spacer length b , proving Theorem 1.

4 Reproducibility and scope

The identity and induction use no computation. A companion Python program, using only exact integer arithmetic and modules from the Python standard library, checks (3) by both polynomial multiplication and the independent pair partition, checks center alignment at every recursive step, and compares the recursive construction with direct multiplication over a bounded regression corpus. From the archive root run

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python3 proof/qanalog_conjecture54_sufficiency.py
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The bounded corpus is regression evidence only; it is not an assumption in the universal proof.

The theorem is a gap-absorption criterion for a product with one lacunary factor. A natural next question is whether there is a two-spacer analogue for

$$\prod_{i=1}^k [a_i]_q [b]_{q^r} [c]_{q^{r'}},$$

with a recursion whose summands again share a common center. Width-four q -Fibonomials already involve two coprime spacer steps, 2 and 3, which is why Theorem 1 does not subsume that case. A two-spacer identity would offer a systematic route toward wider q -Fibonomial coefficients rather than a sequence of unrelated case arguments.

AI assistance. The author used AI assistance in developing the proof presentation and its verification code.

References

- [1] Brendan B. Connelly, Ezekiel Ito, Thomas C. Martinez, Olha Shevchenko, and Kacey Yang. Unimodality of q -Fibonomial coefficients for small cases. *arXiv:2605.12822v1 [math.CO]*, May 12, 2026.